PGCP-AC AUGUST 26

Assignment No. 2

1. Write SQL queries to create Employee and Department tables with suitable columns and constraints.

```
mysql> CREATE TABLE Department (DeptID INT PRIMARY KEY, DeptName VARCHAR(100) NOT NULL, City VARCHAR(50));
Query OK, 0 rows affected

mysql> CREATE TABLE Emp (EmpID INT PRIMARY KEY, EmpName VARCHAR(100) NOT NULL, Salary DECIMAL(10,2), DeptID INT, J
    oiningDate DATE, City VARCHAR(50), FOREIGN KEY (DeptID) REFERENCES Department(DeptID));
Query OK, 0 rows affected
```

2. Write an SQL query to add a new column to an existing table.

```
mysql> ALTER TABLE Emp ADD Email VARCHAR(100);
Query OK, 0 rows affected
Records: 0  Duplicates: 0  Warnings: 0
```

3. Insert at least five records into a table in a single operation.

```
mysql> INSERT INTO Emp (EmpID,EmpName,Salary,DeptID,JoiningDate,City) VALUES
    -> (101,'Tejas',60000,10,'2025-01-15','Mumbai'),
    -> (102,'Khushi',48000,20,'2024-06-10','Pune'),
    -> (103,'Rohan',75000,10,'2023-03-20','Mumbai'),
    -> (104,'Amit',52000,30,'2025-08-01','Delhi'),
    -> (105,'Priya',90000,20,'2022-11-12','Pune');
Query OK, 5 rows affected
Records: 5  Duplicates: 0  Warnings: 0
```

4. Write an SQL query to delete all records from a table.

```
mysql> DELETE FROM Emp;
Query OK, 5 rows affected
```

5. Update salary records from a table where id = 5.

```
mysql> UPDATE Emp SET Salary = 50000 WHERE EmpID = 5;
Query OK, 1 row affected
Rows matched: 1  Changed: 1  Warnings: 0
```

6. Grant SELECT privilege on the Emp table to user readonly_user.

```
mysql> GRANT SELECT ON dbt_assignment2.Emp TO 'readonly_user'@'localhost';
Query OK, 0 rows affected
```

7. Revoke INSERT privilege on the Emp table from user temp_user.

```
mysql> REVOKE INSERT ON dbt_assignment2.Emp FROM 'temp_user'@'localhost';
Query OK, 0 rows affected
```

8. Insert a new employee into Emp and commit the transaction.

```
mysql> START TRANSACTION;
Query OK, 0 rows affected

mysql> INSERT INTO Emp (EmpID,EmpName,Salary,DeptID,JoiningDate,City) VALUES (106,'Sana',55000,10,'2026-10-19','Mu
    mbai');
Query OK, 1 row affected

mysql> COMMIT;
Query OK, 0 rows affected
```

9. Update the salary of employee EmpID = 101 and commit the transaction.

```
mysql> START TRANSACTION;
Query OK, 0 rows affected

mysql> UPDATE Emp SET Salary = 65000 WHERE EmpID = 101;
Query OK, 1 row affected

mysql> COMMIT;
Query OK, 0 rows affected
```

10. Set a SAVEPOINT before updating the DeptID of an employee.

```
mysql> START TRANSACTION;
Query OK, 0 rows affected

mysql> SAVEPOINT before_dept_update;
Query OK, 0 rows affected

mysql> UPDATE Emp SET DeptID = 20 WHERE EmpID = 101;
Query OK, 1 row affected
```

11. Roll back to the previously created SAVEPOINT.

```
mysql> ROLLBACK TO SAVEPOINT before_dept_update;
Query OK, 0 rows affected
```

12. Write an SQL query to truncate all data from the Emp table.

```
mysql> TRUNCATE TABLE Emp;
ERROR 1701 (42000): Cannot truncate a table referenced in a foreign key constraint
```

13. List all employees whose salary is greater than 50,000.

```
mysql> SELECT EmpID, EmpName, Salary FROM Emp WHERE Salary > 50000;
+-------+---------+----------+
| EmpID | EmpName | Salary   |
+-------+---------+----------+
|   101 | Tejas   | 60000.00 |
|   103 | Rohan   | 75000.00 |
|   104 | Amit    | 52000.00 |
|   105 | Priya   | 90000.00 |
+-------+---------+----------+
4 rows in set
```

14. Increase the salary of all employees in DeptID = 10 by 10%.

```
mysql> UPDATE Emp SET Salary = Salary * 1.10 WHERE DeptID = 10;
Query OK, 2 rows affected
Rows matched: 2  Changed: 2  Warnings: 0
```

15. Count the total number of employees in each department.

```
mysql> SELECT d.DeptID, d.DeptName, COUNT(e.EmpID) AS TotalEmployees
    -> FROM Department d LEFT JOIN Emp e ON d.DeptID = e.DeptID
    -> GROUP BY d.DeptID, d.DeptName;
+--------+----------+----------------+
| DeptID | DeptName | TotalEmployees |
+--------+----------+----------------+
|     10 | Sales    |              2 |
|     20 | IT       |              2 |
|     30 | HR       |              1 |
|     40 | Finance  |              0 |
+--------+----------+----------------+
4 rows in set
```

16(a). List departments that have no employees.

```
mysql> SELECT d.DeptID, d.DeptName FROM Department d
    -> LEFT JOIN Emp e ON d.DeptID = e.DeptID
    -> WHERE e.EmpID IS NULL;
+--------+----------+
| DeptID | DeptName |
+--------+----------+
|     40 | Finance  |
+--------+----------+
1 row in set
```

16(b). Create a view for salespeople who belong to Mumbai.

```
mysql> CREATE OR REPLACE VIEW MumbaiSalespeople AS
    -> SELECT EmpID, EmpName, Salary, DeptID, JoiningDate, City
    -> FROM Emp WHERE City = 'Mumbai' AND DeptID = 10;
Query OK, 0 rows affected
```

17. Display the highest salary among all employees.

```
mysql> SELECT MAX(Salary) AS HighestSalary FROM Emp;
+---------------+
| HighestSalary |
+---------------+
|      90000.00 |
+---------------+
1 row in set
```

18. Display the second-highest salary among all employees.

```
mysql> SELECT MAX(Salary) AS SecondHighestSalary
    -> FROM Emp WHERE Salary < (SELECT MAX(Salary) FROM Emp);
+---------------------+
| SecondHighestSalary |
+---------------------+
|            75000.00 |
+---------------------+
1 row in set
```

19. Show all employees ordered by JoiningDate in descending order.

```
mysql> SELECT EmpID, EmpName, JoiningDate FROM Emp ORDER BY JoiningDate DESC;
+-------+---------+-------------+
| EmpID | EmpName | JoiningDate |
+-------+---------+-------------+
|   104 | Amit    | 2025-08-01  |
|   101 | Tejas   | 2025-01-15  |
|   102 | Khushi  | 2024-06-10  |
|   103 | Rohan   | 2023-03-20  |
|   105 | Priya   | 2022-11-12  |
+-------+---------+-------------+
5 rows in set
```

20. Rename the DeptName column to DepartmentName.

```
mysql> ALTER TABLE Department RENAME COLUMN DeptName TO DepartmentName;
Query OK, 0 rows affected
```